

Wellnify Work Logs

Start Date: 14th May, 2024

End Date: 29th October, 2024

Phases	Week	Dates	Task	Description	Hours	Weekly Hours
Project 1: Initial Meetings and Planning	Week 1	14th May, 24 to 19th May, 24	Official Meeting with the Wellnify COO, Mr. Brendon Fergusen and CTO, Mr. Rob Moore	- Introductions ensured all team members and stakeholders were acquainted. - Discussed Wellnify.ai's foundation, goals, and mission. - Reviewed the presentation Wellnify.ai gives to prospective clients. - Provided a comprehensive demo of the current Wellnify app. - Discussed current app limitations and the scope of planned improvements.	4	13.5
			App Overview	- Access to the app was provided. - Spent time individually exploring the app to understand the user perspective. - Noted down any feedback or changes needed. - Gained an understanding of the data structure for analysis.	5	
			Draft an Initial Proposal	- Had a call with team members to discuss potential areas to be explored. - The meeting concentrated on "Data Analysis," "Enhanced Insights," "Web App for displaying new insights," "Upgrades to the existing Dashboard," and "Activity-based Workout Recommendation - Model." - We explored ideas for app enhancements, ensuring they were consistent with Wellnify's brand and addressed their requirements and expectations. - Brainstormed and drafted a proposal for various features and their implementation steps.	4.5	
	Week 2	20th May, 24 to 26th May, 24	Preliminary Project Plan	- Had a meeting with Mr. Rob to review the proposal. - Based on the feedback and requirements gathered in the meeting, we developed a preliminary project plan and divided tasks and roles among the three team members. - All three of us would work on data analysis. Ms. Kritika will focus on the frontend, I will concentrate on the backend, and Mr. Zaid will assist with both.	3	4
			NDA signed	- Read and Signed the NDA for the onboarding process	1	
Project 1: Initial Setup and Exploration	Week 3	27th May, 24 to 2nd June, 24	Initial Database Setup and Exploration	- Gained access to the development database. - Loaded the database into my local workbench environment. - Executed mysql queries to gain an initial understanding of the database and its structure.	4	15
			Database Schema Design	- Implementing database optimizations and indexing	5	
			Challenges in Data Understanding and Analysis	- Encountered challenges in understanding numerous tables and unstructured data in the development database, including a significant amount of garbage values. - Researched methods to understand and analyze the data using Python.	6	
	Week 4	3rd June, 24 to 9th June, 24	Data Structuring	- Utilized Python(Jupyter Notebook) to connect to mysql client and further analyze the data, table structures, and relationships. - Employed Python to clean the data, removing missing or garbage values.	5.5	17.5
			Proposal Reevaluation	- Reevaluated the initial proposal and metrics, assessing feasibility based on the database structure, and devised new metrics to be calculated after gaining a foundational understanding of the database.	5	
			Project Planning and Task Allocation	- Conducted a team call to discuss all deliverables and create a project roadmap. - Split the insights computation and analysis into User and Organization sections. - Developed a plan for creating the web application, selecting the front-end and back-end technology stack. - Researched the best front-end technology and data visualization tools suitable for the project. - Divided the metrics to be calculated among team members, working collaboratively on the analysis.	7	

Project 1: Data Structuring and System Architecture	Week 5	10th June, 24 to 16th June, 24	Weekly Meeting with Mr. Rob and team discussion	- Reviewed the reevaluated metrics proposed by the team. - Clarified any questions and addressed uncertainties regarding the database and its structure. - Received a feedback on the metrics list from Mr. Rob and discussed incorporating new ideas into our proposal.	3	17
			Metric Calculation and Data Preparation	- Divided the calculation of different metrics and features for users and organizations among the three team members. - Identified and merged the necessary tables for analysis, utilizing specific columns to extract required information. - Refreshed knowledge on using Python libraries and statistical methods to calculate new features. Watched tutorials online.	7.5	
			Feature Development and Engineering	- Developed features such as completion rate of user activities, wellness scores, peak active times, and more. - Conducted feature engineering to create new tables and columns for comprehensive data analysis.	6.5	
	Week 6	17th June, 24 to 23th June, 24	System Architecture	- Completed a refresher course on REST APIs using Flask. - Developed an architecture diagram detailing the system setup and established the foundation for the Flask app. - Planned the communication flow between the frontend, backend, and database.	7.5	25
			Weekly Meeting with Mr. Rob	Discussed the metrics derived and addressed concerns regarding data integrity.	1	
			Database Listener Research	- Reviewed and researched methods for setting up listeners on MySQL through Python. - Explored various approaches and best practices for implementing database listeners to monitor changes and trigger actions accordingly.	5	
			Internal Team Discussion	- Based on the meeting outcomes, we made adjustments to our approach, adding new metrics, removing others, and updating our overall plan accordingly. - Reviewed the architecture diagram and data flow, making necessary revisions to the plan.	1.5	
			Designed the new ER Diagram	Designed the database schema for the Wellnify app.- Created ER diagrams and documented relationships between tables.	4	
			Code Integration and Database Setup	- Merged all the code from different notebooks created by various team members. - Pushed the new tables and features to establish our final database.	6	
Project 1:	Week 7	24th June, 24 to 30th June, 24	Weekly Meeting with Mr. Rob	- Provided updates on the project's progress. - Discussed the new features added to our database. - Explored the possibility of working with the staging database to achieve a better data structure.	1	20
			Internal Team Discussion	- Conducted an internal team discussion to further plan the project. - Shared insights on individual problems faced with the data and brainstormed ideas to work around them. - Divided frontend and backend tasks among team members.	2	
			Access and Optimization of Staging Database	- Gained access to the staging database. - Optimized the code to dynamically use the provided URL or configuration of the database host. The code now utilizes the existing data and tables in the database to calculate required features, creating new tables and columns, and then pushing these updates back to the database. - This optimization was implemented to minimize redundant code and reduce the initial setup required when transitioning between local, development, staging, and eventually production databases in the future.	8	
			Access and Optimization of Staging Database	- Transitioned from working with local databases to setting up a database on AWS to ensure synchronicity within the team. - Hosted the final SQL database on AWS for consistent and scalable access.	4.5	
			Proof of Concept Development for Trigger	- Developed a proof of concept to replicate the functionality of database triggers using Flask. - Attempted to populate tables in real-time without using traditional triggers, by scheduling functions written in Jupyter Notebook.	4.5	

Development	Week 8	1st July, 24 to 7th July, 24	Set up the initial Flask environment	- Installing Flask and dependencies. - Configuring the development environment.	5.5	21.5
			Document all the necessary APIs for the project	- Listing and planning API endpoints. - Establish consistent naming conventions and coding standards	5	
			Creating a git Repo	- Setting up the directory layout and configuration files.	4	
			Weekly Meeting with Mr. Rob	- Provided Project Updates	1	
			Internal Team Discussion	- Shared the API Plan with the team and divided APIs to be created between me and Zaid. Kritika will focus on the frontend.	1	
			Updated the Backed Strucuture	- based on the team discussion updated the backend structure	5	
	Week 9	8th July, 24 to 14th July, 24	User API Developement	- Tasked with creating APIs for user-related metrics. - Developed all necessary APIs for user metrics. - Thoroughly tested the APIs to ensure functionality and reliability.	9	24.5
			Feature Enhancement	- Received feedback from the frontend developer to add more user features that could be displayed as charts for providing over-time insights. - Conducted research and brainstorming sessions to explore additional features and metrics that could be calculated for users.	8	
			Bug fixes and Changes	- Perforemed bug fixes based on feedback by frontend developer	5	
			Meeting with Mr. Rob and Internal Team Meeting	- Provided Project updates	2.5	
Project 1: Stakeholder Review and Enhancements	Week 10	15th July, 24 to 21st July, 24	Enhancing User Engagement Analysis	- Added new features and tables for user engagement analysis and integrated these tables into the database. - Modified the notebook code to conduct the initial analysis and store the new features in the tables. - Optimized and cleaned the notebook code for better performance and readability.	7.5	25.5
			Bug fixes and Changes	- Perforemed bug fixes based on feedback by frontend developer	3.5	
			Meeting with Mr. Rob	- Provided Project Updates and discussed about setting up a meeting with Mr. Brendon to give a demo of the app.	1	
			User API Developement	- Added new APIs for the New features and tested the functionality	6.5	
			Internal Team Discussion	- Internal team meeting to go over the entire project and test everything before the meeting with the stakeholders.	2.5	
			Bug fixes and Changes	- Bug fixes and changes based on the meeting.	4.5	
	Week 11	22nd July, 24 to 28th July, 24	Meeting with the Stakeholders	- Held a meeting with Mr. Rob and Mr. Brendon to give a comprehensive demo of the entire app. - Discussed potential enhancements to the metrics and how they are calculated. - Explored how the new features and metrics would be integrated into their existing system. - Outlined future steps for continued development and improvement of the app.	2	10
			Code Enhancement and Integration	- Restructured the code in the Jupyter Notebook to enhance readability. - Added logging to the Jupyter Notebook scripts according to industry standards. - Combined the notebook code and backend code into a single repository. Encountered and resolved multiple errors while doing so - Implemented logic in the backend to integrate and execute the notebook scripts.	8	
	Week 12	29th July, 24 to 4th Aug, 24	Created Docker File	- Set up Docker configurations for the Jupyter Notebook to ensure consistency across development environments. - Created a Dockerfile to manage dependencies and simplify deployment.	6.5	25.5
			Resolved Errors in the Trigger	- Identified and resolved specific errors in the database trigger implementation. - Conducted tests to verify the trigger functionality and ensure data consistency.	4	
			Meeting with Mr. Rob to Discuss Triggers	- Discussed issues related to triggers and explored possible solutions. - Received feedback on current progress and suggestions for improvement.	1	
			Internal Team Discussion	- After evaluating the execution speed and scalability, decided to transition from Jupyter Notebooks to standalone Python scripts.	2.5	

			Created Python Scripts	- Developed Python scripts to replicate and enhance the data analysis functions previously implemented in the Jupyter Notebook. - Simplified the notebook by transferring core processing tasks to the scripts. - Verified script accuracy and documented the changes.	6.5	
			Updated the Docker files	- Created individual Docker files for each Python script to automate execution via daily cron jobs. - Ensured that each Docker container executed the script as scheduled, maintaining data processing consistency. - Tested the cron job functionality to verify proper daily execution.	5	
	Week 13	5th Aug, 24 to 11th Aug, 24	Performed Comprehensive Testing	- Conducted comprehensive testing of the dashboard interface, Python scripts, data analysis processes, and overall system functionality. - Validated data accuracy, script performance, and dashboard responsiveness to ensure a seamless user experience. - Documented test results and any issues identified.	6	14.5
			Internal Team Discussion	- Held an internal team meeting to review project status, discuss final adjustments, and confirm completion of all deliverables. - Agreed on next steps for final documentation and prepared for stakeholder review.	2	
			Bug fixes and Changes	- Addressed and resolved bugs identified during testing to ensure system stability and functionality. - Conducted follow-up tests after each fix to confirm resolution and verify performance improvements.	5	
			Meeting with the Stakeholders	- Presented final project outcomes, highlighted key accomplishments, and discussed overall impact. - Engaged in a wrap-up discussion, covering lessons learned and insights for potential future improvements. - Explored ideas for upcoming projects and ways to expand Wellnify's capabilities.	1.5	
	week 14	12th Aug, 24 to 18th Aug, 24	Discuss Next Project Ideas	- Held an in-depth brainstorming session with the team to discuss potential project directions. - Evaluated each idea based on feasibility, potential impact, and alignment with Wellnify's mission. - Reviewed recent trends in recommendation systems to identify high-impact areas for user engagement. - Documented key takeaways and outlined initial thoughts on each project idea.	2.5	21
			Explored Options for Recommendation System: LLM-Based vs. ML-Based Approaches	- Conducted a detailed exploration of both LLM-based and ML-based recommendation systems, assessing technical requirements for each. - Created a comparison matrix highlighting pros, cons, resource demands, and estimated timelines for each approach. - Drafted a preliminary outline for how each method could be integrated into Wellnify's existing data and architecture. - Held a follow-up internal discussion to refine the comparison and address questions.	5.5	
			Conducted Research on LLM-Based and ML-Based Recommendation Systems	- Researched current best practices, focusing on how each recommendation approach performs with large datasets and various data types. - Investigated relevant tools and libraries for each approach (e.g., Transformers library for LLMs, Scikit-Learn, and TensorFlow for ML-based approaches). - Analyzed case studies from similar wellness or engagement platforms to understand practical implementation challenges. - Compiled a list of potential datasets and additional features required for each approach, with emphasis on availability and compatibility.	7.5	
			Meeting with Mr. Rob to Decide on the Recommendation System Approach	- Presented findings from research to Mr. Rob, highlighting the strengths and weaknesses of both approaches. - Based on project goals and resources, collectively decided to proceed with the ML-based recommendation system for initial implementation.	1.5	

Project 2: Initial Meetings and Planning			Created Documentation for Project Requirements and Initial Ideas	- Drafted comprehensive documentation capturing research findings, project scope, and technical requirements. - Organized detailed notes on the ML-based recommendation system, covering model design, anticipated challenges, and dependencies. - Added a section on preliminary feature engineering ideas for the recommendation model, including initial data processing requirements. - Created a visual roadmap to illustrate the project timeline and key phases.	4	
	Week 15	19th Aug, 24 to 25th Aug, 24	Developed Project Plan	- Structured the project plan, including specific tasks, deliverables, and timelines. - Assigned preliminary responsibilities to each team member, aligning expertise with project needs. - Outlined milestones and checkpoints for tracking progress, including regular review sessions. - Incorporated risk management strategies, identifying potential bottlenecks and contingency plans. - Drafted initial testing and validation strategies for model evaluation and potential iteration phases.	4.5	25
			Content-Based vs. Collaborative Filtering for Recommendation System	- Evaluated the strengths and weaknesses of content-based recommendation versus collaborative filtering, focusing on factors like data availability, personalization, and scalability. - Documented how each approach aligns with Wellnify's goals, particularly regarding user engagement and system compatibility. - Prepared a summary with potential implementation pathways and shared it in the internal team space for feedback.	4.5	
			Meeting with Mr. Rob	- Presented the findings to Mr. Rob, discussing how each approach aligns with Wellnify's objectives. - Mr. Rob advised proceeding with the content-based recommendation model due to its suitability for Wellnify's current data structure and user engagement goals. - Established immediate tasks, such as feature extraction and model design, to initiate development of the content-based system.	1.5	
			Review and Analysis of Latest Production Database	- Accessed and examined the latest production database to assess available data fields and relevant attributes for the content-based recommendation system. - Conducted a detailed review of user behavior data, content tags, and engagement metrics to identify potential features for the recommendation algorithm. - Created a data map detailing key tables, relationships, and fields that would be crucial for model training and recommendation scoring.	6	
			Research and Tutorials on Content-Based Recommendation Implementation	- Browsed through tutorials, documentation, and case studies to explore methods for implementing a content-based recommendation system. - Focused on understanding practical applications of content-based filtering, including feature engineering, similarity measures (e.g., cosine similarity), and model evaluation techniques. - Documented insights and best practices applicable to the Wellnify use case, including suggestions on scalability and optimization techniques. - Experimented with small-scale code snippets to test basic implementation approaches and assess feasibility.	6.5	
			Internal Team Discussion	- Convened an internal team meeting to share findings from the database review and tutorial research. - Discussed initial steps for feature extraction and similarity metric selection, ensuring alignment on project direction. - Outlined upcoming tasks and responsibilities for each team member, setting milestones for the development phase. - Addressed any initial concerns or questions regarding data handling and model design.	2	
			Meeting with the Stakeholders	- Outlined data extraction and discussed essential data elements for the recommendation model, such as collections, tag mappings, and activity links.	1.5	
			Team Discussion	- Reviewed data validation, planned handling of null values, and defined tasks for structured data extraction.	1	

Project 2: Data Preperation	Week 16	26th Aug, 24 to 1st Sep, 24	SQL Query Development for Data Extraction	- Created SQL queries to extract key data fields for the recommendation model, ensuring that all necessary fields (such as tag mappings and activity links) were included. - Tested queries on sample tables, validating initial data pulls by manually checking row counts and key field values to ensure accuracy. - Documented each query's purpose and parameters for future reference, ensuring any changes could be tracked and reproduced. - Developed additional queries to extract supplementary fields, such as timestamps and activity levels, to be used in later stages of model refinement. - Implemented exception handling for SQL scripts to manage potential errors or missing data scenarios, logging issues for follow-up.	8	26
			Data Cleanup and Null Handling	- Applied pandas-based data cleaning functions to handle missing values in critical fields, including conditional replacements for common gaps in tags or activity details. - Created a process to identify and flag duplicate records across tables, logging the frequency and types of duplicates for analysis. - Standardized text fields (e.g., tags, collection names) by converting to lowercase and replacing spaces with underscores to ensure consistency in similarity calculations. - Verified the consistency of cleaned data by cross-referencing a subset of records with the original database entries, ensuring no critical information was lost in the cleanup. - Documented the data cleaning steps for reproducibility and future updates, including any edge cases handled during processing.	9	
			Initial Data Storage Setup	- Organized the cleaned and validated data into structured CSV files, categorizing each dataset by its role in the recommendation pipeline (e.g., primary features, auxiliary features). - Saved metadata for each file (e.g., row count, data fields) to support data lineage tracking and facilitate future debugging. - Conducted initial data profiling for each CSV file to summarize row counts, missing values, and unique entries, capturing this in a report to guide future feature engineering.	6.5	
			Meeting with Mr. Rob	- Presented extracted data samples, highlighting the key fields and structure prepared for feature engineering. - Discussed potential transformations, including encoding methods for categorical data and handling of hierarchical tags and activity categories. - Took notes on feedback related to future-proofing the dataset for additional features, ensuring scalability in data transformations. - Created a follow-up action plan to address specific data transformation requirements discussed during the meeting	1.5	
			Internal Team Discussion	- Held a team-wide discussion to finalize encoding strategies for different types of features, including tags, activities, and collection types. - Explored potential encoding options beyond one-hot encoding, such as binary and frequency encoding, for fields where one-hot encoding may add too many columns. - Drafted a shared guide on encoding standards and procedures to maintain consistency in encoding across all data features. - Assigned individual responsibilities for encoding tasks to ensure timely completion, with each team member focusing on specific categories of features.	2	
			One-Hot Encoding for Categorical Data	- Applied one-hot encoding to tags, collection types, and activities, creating a binary matrix for each category and verifying that all unique categories were included. - Ran tests on the encoded matrices, cross-checking the original categories with the one-hot encoded columns to ensure no data loss or inconsistencies. - Resolved any errors encountered during encoding, such as overlapping category names, by creating unique identifiers for each encoded feature. - Documented the entire encoding process for each feature type, noting any limitations and challenges for future reference.	8	

Project 2: Feature Engineering and Transformation	Week 17	2nd Sep, 24 to 8th Sep, 24	Multi-Label Binarization for Multi-Category Tags	- Used MultiLabelBinarizer to encode features with multiple tag associations, creating a binary matrix that captures each possible combination of tags. - Verified that each tag combination was correctly represented in the binary matrix, ensuring complete coverage of multi-label associations. - Conducted tests to verify the accuracy of the multi-label binarization and the compatibility of this matrix with other encoded features. - Adjusted binarization settings to improve performance and accommodate future additions to the tag categories.	8	39.5
			Consolidation of Encoded Features	- Combined all one-hot encoded and multi-label binarized features into a single dataset, creating a comprehensive feature matrix for model training. - Checked for potential redundancies in columns resulting from overlapping tags or categories and removed duplicate entries. - Created and saved multiple versions of the consolidated dataset to allow experimentation with different combinations of features. - Added metadata to each version of the consolidated feature matrix, including details on column count, feature types, and matrix size.	6	
			Feature Scaling and Normalization	- Applied Min-Max scaling to numerical fields to standardize feature values between 0 and 1, improving compatibility with similarity calculations. - Tested normalization approaches for each category, such as using Z-score normalization for specific columns, ensuring consistent value ranges across features. - Validated the scaled data to check for any anomalies, such as values outside the expected range, correcting as needed. - Documented the scaling and normalization steps, specifying the approach taken for each type of feature and the rationale behind it.	5.5	
			Standardization of Encoded Data for Similarity Calculations	- Standardized all encoded and normalized features, converting them into consistent formats for compatibility across various similarity functions. - Ensured uniform distribution and unit variance across all feature columns to prevent bias toward certain encoded features. - Conducted further testing to ensure standardized features aligned with the requirements for cosine similarity calculations.	5	
			Documentation	- Compiled a comprehensive report on the data transformation and encoding process, including challenges encountered and resolutions. - Created visual diagrams to illustrate the encoding and scaling processes for each feature type, enhancing understanding for future project phases. - Organized documentation for easy reference, enabling other team members to access a detailed outline of the transformation workflow.	3.5	
			Meeting with the Stakeholders	- Discussed the application of TF-IDF to convert text-based content into numeric form, capturing relevance for each term in collections and tags. - Reviewed alternative text processing techniques like word embeddings and evaluated the suitability of TF-IDF based on the current data structure	1.5	
			Internal Team Discussion	- Conducted a team session to finalize workflows for text preprocessing, ensuring the inclusion of all relevant content fields (e.g., descriptions, summaries). - Drafted a document to guide the workflow, specifying text preprocessing steps, including tokenization, stop word removal, and lemmatization. - Assigned roles for each stage of text processing and established a timeline for completing these tasks.	2	
			Tokenization and Text Cleaning	- Tokenized each content summary field, splitting text into individual words for further processing. - Removed stop words and non-informative characters, standardizing terms across documents to improve vectorization results. - Verified consistency in cleaned text data, ensuring that only meaningful terms were retained. - Conducted tests to ensure that the cleaned tokens matched the original content summaries in intent and information.	7.5	

	Week 18	9th Sep, 24 to 15th Sep, 24	TF-IDF Vectorization for Text Data	- Implemented TF-IDF vectorization using Scikit-Learn to transform content summaries into numerical form, focusing on term frequency and inverse document frequency. - Conducted experiments with different parameters in the TF-IDF model, such as max_df, min_df, and ngram_range, to optimize the importance of terms. - Generated sparse matrices representing the importance of each word in the context of each collection, capturing key content relevance. - Compared various vectorized matrices to identify the configuration that best represented content relevance for the recommendation system.	8.5	43.5
			Dimensionality Reduction for TF-IDF Matrix	- Applied Singular Value Decomposition (SVD) to reduce the dimensionality of the TF-IDF matrix, enhancing memory efficiency without sacrificing relevance. - Tested different numbers of components for SVD, balancing dimensionality with accuracy in text representation. - Verified that the reduced matrix retained core semantic information, essential for similarity calculations in the recommendation model. - Documented the SVD process, including the optimal number of components used for dimensionality reduction.	7	
			Sparse Matrix Optimization and Consolidation	- Combined the TF-IDF matrix with previously one-hot encoded categorical features using SciPy's hstack to create a single sparse matrix. - Verified alignment between the combined sparse matrix columns, ensuring consistent representation across features. - Applied memory optimization techniques to manage storage requirements, such as using efficient data types for matrix values. - Documented the sparse matrix structure, specifying the purpose of each feature group within the consolidated matrix.	8	
			Testing and Validation of Text-Based Feature Matrix	- Conducted validation tests to confirm the accuracy of each vectorized feature in the sparse matrix, checking consistency with the original data. - Ran multiple test cases to ensure the matrix provided stable and accurate values for text-based features across various collections. - Created and executed unit tests for individual feature sets within the matrix, documenting results and noting any edge cases for future reference.	5.5	
			Documentation	- Compiled comprehensive documentation detailing each stage of the text processing workflow, from tokenization to matrix consolidation. - Prepared a report on the optimization techniques used to enhance the TF-IDF and sparse matrix creation, including trade-offs in dimensionality reduction.	3.5	
			Meeting with the Stakeholders	- Presented an overview of potential similarity calculation techniques, focusing on cosine similarity and weighted feature scoring. - Discussed approaches for feature weighting, particularly for categorical data, to emphasize certain attributes (e.g., tags or activities) based on user engagement. - Reviewed use cases for multi-feature recommendations, addressing how combined feature weights would improve personalized recommendations. - Documented Mr. Rob's feedback on model priorities, setting a foundation for iterative improvement in future development.	2	
			Internal Team Discussion	- Conducted a team meeting to finalize the modular approach for calculating similarity on each feature type, ensuring flexibility for future feature additions. - Discussed potential benefits and drawbacks of different similarity metrics, such as Euclidean distance and Jaccard similarity, ultimately reaffirming cosine similarity as the preferred approach.	2.5	

Project 2: Similarity Calculation and Model Tuning	Week 19	16th Sep, 24 to 22nd Sep, 24	Cosine Similarity Calculation	- Implemented cosine similarity for individual features (tags, collection types, and activities) using Scikit-Learn's cosine_similarity function. - Fine-tuned cosine similarity parameters based on preliminary test results, ensuring similarity scores effectively represented user content preferences. - Built modular functions to calculate similarity independently for each feature, simplifying maintenance and future adjustments. - Verified feature-level similarities by checking consistency with raw data, ensuring that scores aligned with expected outcomes.	8	38.5
			Weighted Feature Combination for Multi-Feature Similarity	- Developed a system to combine individual feature similarity scores, allowing adjustable weights for each feature type. - Experimented with different weight distributions, testing various combinations to find an optimal balance for each feature's influence. - Built a scalable weighting mechanism, allowing easy adjustments to feature weights based on future insights or changes in recommendation priorities. - Documented all configurations tested and the rationale behind the chosen weight distribution.	7.5	
			Testing and Adjustments of Initial Similarity Scores	- Conducted test runs on similarity matrices across different user profiles, evaluating how well the model captured relevant recommendations. - Analyzed test results to identify any biases in feature representation, particularly focusing on high-frequency tags that might dominate recommendations. - Adjusted feature weights and similarity scoring parameters to mitigate biases, iterating through multiple configurations. - Compiled testing results and documented findings, noting the most effective parameter adjustments for future use.	7	
			Created and Validated Combined Similarity Matrix	- Combined individual feature-specific similarity matrices into a single, unified similarity matrix representing overall similarity scores. - Validated the combined matrix by comparing scores with baseline test cases, ensuring consistency and reliability of results. - Built and ran validation scripts to automate consistency checks for the similarity matrix, identifying any discrepancies for further adjustment. - Saved multiple versions of the combined similarity matrix with various weight configurations to enable future comparisons.	6	
			Documentation and Report on Similarity Calculations	- Compiled a detailed report documenting the similarity calculation process, feature weight tuning, and testing results. - Created visual aids (e.g., charts and diagrams) to illustrate the impact of feature weighting on recommendation accuracy. - Documented the rationale for each weighting configuration tested, providing a clear record for future tuning or feature additions. - Presented the documentation to the team for review and incorporated feedback to enhance clarity and completeness.	5.5	
			Meeting with Mr. Rob to Discuss Weight Tuning	- Presented initial similarity scores for each feature, discussing how weights impacted overall recommendation relevance. - Reviewed user feedback and engagement data to identify patterns that could inform weight adjustments. - Discussed ideas for dynamically adjusting weights based on user behavior and preferences. - Outlined an approach to balance tag, activity, and content-based similarities, incorporating Rob's feedback for iterative refinement.	2	
			Brainstorming on Weight Adjustment Techniques	- Explored techniques such as incremental weight adjustment, normalization, and dynamic scaling based on feature impact. - Analyzed potential issues, like over-emphasis on frequently occurring tags, and brainstormed solutions for balanced recommendations. - Created a reference guide on weight adjustment strategies, including examples of weight distributions for different user scenarios.	4	

	Week 20	23rd Sep, 24 to 29th Sep, 24	Iterative Weight Adjustment and Testing	- Implemented an iterative tuning process, adjusting weights for each feature (TF-IDF, tags, and activities) across multiple test scenarios. - Ran tests to observe the impact of each configuration on recommendation quality, recording the results of each test in detail. - Used a grid search method to systematically evaluate different weight combinations, identifying those that provided the highest engagement. - Documented insights from testing, including optimal configurations and adjustments that enhanced recommendation relevance.	9	40.5
			Integration of Smoothed Inverse Frequency for Tag Weighting	- Applied a smoothed inverse frequency method to re-weight tags, reducing the influence of frequently occurring tags and emphasizing unique or niche tags. - Tested different smoothing factors to balance the impact of common tags, ensuring that rare tags contributed to personalized recommendations. - Validated the inverse frequency adjustments across various user profiles, confirming improved personalization and diversity in recommendations. - Documented the process of applying inverse frequency adjustments and saved multiple configurations for comparison.	7.5	
			User-Centric Weight Tuning and Adaptive Recommendations	- Conducted additional tests with weights tailored for specific user profiles, focusing on personalized recommendations based on user interaction data. - Built and tested adaptive weighting methods that adjusted based on user engagement patterns, providing tailored content recommendations. - Recorded findings and insights from each profile-based test, helping to refine the weight tuning approach for user-specific recommendations.	7	
			Testing and Validation of Adjusted Weight Configurations	- Ran extensive validation tests to confirm the stability and accuracy of the final weight configurations across different data sets. - Created automated test cases to validate weight adjustments in new data, ensuring consistency and accuracy in ongoing model updates. - Compiled validation results to highlight the impact of weight tuning on recommendation quality, with detailed explanations of observed improvements.	6.5	
			Documentation of Weight Tuning Process and Final Configuration	- Compiled a detailed report documenting each step of the weight tuning process, including the rationale for selected configurations. - Documented the final weight distribution, including the influence of each feature and the adjustments made to improve recommendation relevance. - Created a reference guide with recommended configurations for future adjustments, enabling easy replication and customization.	4.5	
			Meeting with Mr. Rob on API Design	- Reviewed the desired API structure, identifying critical endpoints for real-time recommendation delivery. - Discussed scalability requirements, considering high user loads and frequent requests, and outlined solutions for handling peak demand. - Explored potential caching strategies to minimize response times and reduce server load during heavy API usage. - Gathered feedback on data security measures, emphasizing user privacy and secure data handling.	2	
			Internal Team Discussion	- Conducted a collaborative session to finalize endpoint specifications, security protocols, and error-handling strategies. - Outlined endpoint requirements, including HTTP methods, parameters, and data formats for seamless integration. - Developed a security plan for API access, discussing options such as token-based authentication and rate limiting to prevent abuse. - Documented potential errors, defining clear error messages and status codes for easier debugging and user feedback.	2	

Project 2: API Development	Week 21	30th Sep, 24 to 6th Oct, 24	Implementation of Recommendation API Endpoints	- Developed multiple Flask-based endpoints for recommendation delivery, including endpoints to retrieve recommendations by user ID and filter by category or tags. - Integrated query parameters to allow customization, such as specifying the number of recommendations or filtering by content type. - Implemented caching mechanisms to optimize response times for frequently requested recommendations. - Conducted initial tests on each endpoint to ensure functionality, handling both successful responses and error cases effectively.	8	37
			Error Handling and Logging for API Endpoints	- Implemented comprehensive error-handling logic to manage various scenarios, including invalid requests, data retrieval issues, and server errors. - Added logging at critical points within each endpoint to capture request details, response times, and any encountered errors for future troubleshooting. - Created standardized error messages and status codes, improving user experience by providing clear and actionable feedback. - Documented logging configurations and error-handling protocols to facilitate maintenance and debugging.	6	
			Integration Testing of API with Model and Data Pipeline	- Conducted integration tests, simulating end-to-end API requests to ensure smooth interaction between the API, recommendation model, and data pipeline. - Validated API responses for accuracy, consistency, and speed, running tests on various devices and network conditions. - Used load-testing tools to simulate high traffic, measuring API response times and identifying potential bottlenecks. - Adjusted configurations to optimize performance, based on testing results, ensuring real-time responsiveness even under peak load.	7.5	
			Security Testing and Authorization Configuration	- Configured token-based authentication to secure API endpoints, restricting access to authorized users only. - Performed security tests to verify that sensitive data remained protected, simulating potential attack vectors to ensure API robustness. - Implemented additional security layers, including encryption for data in transit and rate limiting to prevent abuse. - Created guidelines for secure API access, including recommendations for token storage and access control.	6	
			API Documentation	- Developed comprehensive documentation for each API endpoint, specifying input parameters, response structures, and expected error messages. - Created a user-friendly guide on API setup, with examples demonstrating common usage scenarios and customization options. - Documented security protocols and best practices for API use, providing guidance on authorization, authentication, and data handling. - Reviewed documentation with the team to ensure clarity, addressing any questions or feedback for refinement.	5.5	
			Meeting with Mr. Rob	- Discussed requirements for retrieving recommendations dynamically, specifically for organization-specific and user-specific contexts. - Reviewed potential use cases for personalized recommendations based on organizational filters, aiming to increase engagement within user groups. - Addressed the impact of dynamic retrieval on API performance, discussing options to balance personalized results with response times. - Outlined criteria for evaluating the relevance of recommendations in both user-specific and organization-specific scenarios.	2	

	Week 22	7th Oct, 24 to 13th Oct, 24	Discussion on Filter Criteria for Dynamic Recommendations	- Conducted a brainstorming session to determine effective filtering criteria based on user preferences, engagement history, and organizational affiliation. - Finalized filter parameters, such as age, activity level, and preferred content types, to ensure that recommendations align with user-specific interests. - Discussed how to prioritize certain filters in recommendations, balancing relevance with personalization.	3	40
			Developed Organization-Based Filtering in API	- Integrated organization-based filters into the API, allowing recommendations to be tailored according to the user's organizational context. - Implemented logic to check user affiliations and filter content based on the organization's preferred or relevant collections. - Ensured that organization-based filters dynamically adjust recommendations, maintaining relevance as organizational data changes. - Conducted tests on organization-based filtering with sample data from various organizations, verifying alignment with specific organizational themes.	8.5	
			User-Specific Filtering and Recommendation Customization	- Developed user-specific filters, including preferences based on past interactions, activity type, and engagement levels. - Designed a system to update recommendations in real-time based on recent user interactions, personalizing content suggestions dynamically. - Tested user-specific filtering across multiple profiles, ensuring that recommendations accurately reflected unique user preferences and behaviors. - Documented the customization logic for user-specific recommendations, allowing for future enhancements based on user feedback.	8	
			Recommendation Logging for Interaction Tracking	- Established a logging system to track user interactions with recommendations, capturing key metrics such as click-through rates, time spent on content, and user feedback. - Configured logs to categorize data by interaction type, making it easy to analyze trends in user engagement across different recommendation types. - Implemented log rotation to manage storage efficiently, ensuring logs are retained for analysis without overloading storage capacity. - Verified that all essential interactions were logged, confirming that the data could support in-depth analysis of recommendation performance.	7.5	
			Initial Analysis of Logged Interaction Data	- Conducted preliminary analysis on the logged data to identify patterns in user engagement, focusing on high-interest content categories and frequent user preferences. - Used statistical methods to measure engagement trends, such as average interaction time, click-through rates, and user retention. - Compiled a report summarizing key engagement insights, noting popular recommendation filters and any discrepancies in interaction patterns.	6	
			Documentation and Reporting on Filtering and Logging Systems	- Documented the filtering logic for both organization-based and user-specific recommendations, providing examples for future reference and maintenance. - Compiled a detailed guide on the logging system, explaining its setup, data captured, and usage for tracking recommendation performance. - Created a summary report on initial findings from the interaction data analysis, highlighting areas of interest and proposed next steps for tuning the model. - Reviewed documentation with the team to incorporate feedback, ensuring clarity and completeness.	5	
			Meeting with Mr. Rob	- Presented an overview of interaction logging data collected, highlighting key metrics such as engagement rates and time spent on recommended content. - Discussed initial insights derived from user behavior patterns and gathered feedback on areas of the model that could benefit from refinement.	2	

Project 2: Model Refinement, Demo Preparation, and Project Handoff	Week 23	14th Oct, 24 to 20th Oct, 24	Internal Team Review of User Interaction Data and Logging Consistency	- Conducted a team-wide review of the logging process, identifying any inconsistencies in captured data or gaps in interaction tracking. - Aligned on adjustments to improve logging consistency, ensuring that all relevant interactions were uniformly recorded across user profiles. - Established a protocol for verifying the completeness and accuracy of logged data, setting up regular checks to maintain logging standards.	2.5	37.5
			User Feedback Analysis and Model Adjustment	- Conducted an in-depth analysis of user feedback data to identify high-engagement content types, examining which recommendations resonated most with users. - Analyzed feedback to detect patterns related to specific tags, content types, or activity levels, gaining insights into user preferences. - Adjusted the model to prioritize high-engagement content types, refining the recommendation algorithm to align with observed user interests. - Documented adjustments made to the model and compiled insights from the analysis for future model refinement sessions.	7.5	
			A/B Testing Setup for Model Configuration Evaluation	- Designed and implemented A/B testing to evaluate the effectiveness of different model configurations, comparing user engagement across multiple setups. - Created distinct test groups for each configuration, focusing on variations in feature weights, similarity metrics, and content prioritization. - Configured data logging for each A/B test to capture metrics on engagement, retention, and user satisfaction, providing a basis for comparative analysis. - Ran initial tests to ensure accurate tracking of results, confirming that each configuration delivered distinct recommendation outcomes.	7	
			Iterative Refinement Based on A/B Testing Result	- Analyzed A/B testing results to determine which configuration yielded the highest engagement and relevance, focusing on click-through rates and session duration. - Refined model configurations based on insights from the tests, making iterative adjustments to weights and similarity scores. - Conducted follow-up tests to verify that adjustments improved model performance, fine-tuning the recommendation system for maximum relevance. - Compiled a report on A/B testing findings, outlining the rationale for final adjustments and the impact on user engagement.	6	
			Implementation of Continuous Improvement Loop	- Set up a continuous improvement loop by integrating user feedback and engagement data directly into the model tuning process, enabling ongoing refinement. - Established procedures for regular review of interaction data, ensuring the model evolves with changing user preferences and trends. - Automated certain aspects of the feedback loop, such as collecting and analyzing engagement metrics on a scheduled basis, to streamline the improvement process. - Documented the continuous improvement framework, including guidelines for tracking performance, reviewing insights, and applying model updates.	7	
			Documentation on Model Refinement	- Created comprehensive documentation of the interaction tracking and A/B testing process, detailing setup, metrics tracked, and insights gained. - Compiled a report summarizing the continuous improvement loop, providing a structured approach for regular model updates based on user behavior. - Included visual representations of engagement data, A/B testing results, and model adjustment impacts for clearer communication of findings. - Reviewed the documentation with the team and incorporated feedback to ensure it was comprehensive and aligned with future goals.	5.5	
			Meeting with the Stakeholders	- Reviewed the structure of the demo and outlined essential metrics to showcase, such as engagement rates and recommendation relevance. - Discussed key scenarios and features to highlight, ensuring the demo aligned with stakeholder interests. - Finalized Rob's feedback on specific metrics and features to emphasize for a clear demonstration of the model's impact.	1.5	

	Week 24	21st Oct, 24 to 27th Oct, 24	Team Demo Rehearsal and Final Adjustments	- Conducted a full rehearsal, gathering feedback from the team on flow and user experience. - Made necessary adjustments based on feedback, refining visuals, transitions, and explanations to improve clarity and impact.	3	15.5
			User Interface Enhancements for Demo Readiness	- Updated the UI to enhance presentation quality, adding tooltips and brief descriptions to guide users through the recommendation features. - Performed tests to confirm compatibility and responsiveness, ensuring a seamless presentation. - Documented UI changes for future improvements.	4	
			Performance Optimization for Real-Time Demo	- Fine-tuned backend processes to support real-time interactions during the demo, focusing on response time and efficiency. - Ran stress tests to confirm system stability and responsiveness under simulated high-traffic conditions.	3	
			Final Model Tuning for Enhanced Relevance	- Conducted final adjustments on feature weights and personalization settings based on recent feedback, improving recommendation relevance. - Tested the model's performance across varied user profiles, verifying consistent quality and relevance.	4	
	Week 25	25th Oct, 24 to 29th Oct, 24	Final Meeting with Mr. Rob and Mr. Brendon for Final Project Review and Handoff	- Conducted a comprehensive wrap-up meeting, presenting the project's final outcomes, highlighting system performance metrics, and discussing areas for future development. - Gathered feedback from Mr. Rob and Mr. Brendon, noting key insights and suggested improvements for ongoing model refinement. - Outlined next steps for system maintenance, ensuring alignment on goals for continued monitoring and potential expansions.	2	18.5
			Internal Team Meeting to Review Documentation and Integration	- Held a final internal review to verify that all documentation was complete, accurate, and accessible to team members and future collaborators. - Reviewed integration points, ensuring that the handoff process was seamless and that onboarding materials were readily available. - Addressed any last-minute questions from team members about system components, establishing a smooth transition for future maintenance.	2	
			Comprehensive Documentation of System Architecture and Codebase	- Compiled detailed documentation for the entire codebase, covering API specifications, data pipelines, recommendation model setup, and deployment configurations. - Created clear diagrams and flowcharts illustrating the system's architecture, data flow, and integration points, ensuring ease of use for future developers. - Organized documentation into structured sections, making it easily navigable and suitable for various audiences, from developers to stakeholders.	5.5	
			Final Quality Assurance and Testing	- Performed a final quality assurance check across the system, including API response times, recommendation accuracy, and logging functionality. - Conducted tests under various scenarios to confirm system stability and responsiveness, documenting any minor issues or adjustments needed. - Compiled a summary report of the quality assurance results, including recommendations for periodic testing routines and maintenance tasks.	5	
			Project Handoff Documentation and Report	- Created a concise project handoff document summarizing key project components, outcomes, and areas for future development. - Included links to detailed documentation, resources, and contacts for easy access by future developers or team members. - Reviewed the handoff report with the team, ensuring all sections were covered and that it provided a complete overview of the system's current state.	4	